# AYRA AGRAWAL

New Brunswick, NJ
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## Education

**Rutgers University – New Brunswick** Expected May 2029
*B.S. Computer Science | B.A. Psychology | Minor: Data Science | GPA: 3.6/4.0* *New Brunswick, NJ*
**Relevant Coursework:** Data Structures, Computer Architecture, Discrete Structures I, Linear Algebra (in progress)

## Technical Skills

**Languages**: Python, Java, C, JavaScript/TypeScript, SQL, R, HTML/CSS
**Frameworks & Data**: FastAPI, SQLAlchemy, Next.js, React, React Native, Node.js, PostgreSQL, pgvector, Redis
**DevOps & Testing**: Docker, AWS EC2, Vercel, Supabase, GitHub Actions (CI), Pytest, Vitest, Playwright, Git
**AI & Research Tools**: Ollama, Sentence Transformers, MediaPipe, Gemini API, PsychoPy, Tobii Pro, DeTox

## Experience

**Frontend Software Engineering Fellow** Oct 2026 – Present
*Rutgers Blueprint* *New Brunswick, NJ*

**Backend Developer** Sept 2026 – Present
*Hack4Impact – Rutgers Chapter* *New Brunswick, NJ*
- Building Supabase backend systems (database, auth, real-time updates, file storage) for a cross-chapter app for nonprofit Love Blue Inc., integrating Google Maps and Zeffy to replace manual beach-cleanup tracking.
- Designed role-based access so volunteers, chapter leaders, and admins can log cleanups and track rankings.

**Undergraduate Research Assistant & IT Team Member** May 2026 – Present
*Cognition and Learning Center (CALC), Rutgers Center for Cognitive Science (RuCCS)* *New Brunswick, NJ*
- Engineered R pipelines extracting measurement periods, frame onsets, and looking-duration metrics for an infant study, with automated QA validation flagging inconsistencies across 227+ coding files (∼300 datasets).
- Configured, tested, and maintained Tobii eye-tracking hardware, PsychoPy, and DeTox to ensure reliable, standardized data collection for ManyBabies, an 800+ researcher, 50+ country consortium.
- Building EyeSpy, a PsychoPy number-word study; coded infant gaze in Datavyu and analyzed SFON counting data.

**Backend Developer** Feb 2026 – May 2026
*Rutgers University Mobile App Incubator* *New Brunswick, NJ*
- Engineered backend systems for RUThrift, a Rutgers-exclusive marketplace app, handling auth, profiles, listings, and messaging with Supabase and PostgreSQL; weighed OTP vs. SMTP verification for onboarding speed and security.
- Shipped the app end-to-end on a 4-person team (React Native, Expo, Git) through a competitive incubator and presented it live at the RUMAD showcase to 50+ attendees.

## Projects

**TraceLab – Distributed Tracing & Observability Backend** | *GitHub*
*Python, FastAPI, PostgreSQL, SQLAlchemy, Redis, Docker, AWS EC2, GitHub Actions, Prometheus*
- Engineered a FastAPI backend to ingest, query, filter, and paginate distributed traces, modeling parent-child span hierarchies in PostgreSQL via SQLAlchemy; deployed a four-service Docker Compose stack to AWS EC2.
- Built async Redis-queue ingestion with a background worker, idempotent writes, and Prometheus metrics.
- Load-tested async ingestion with Locust at 100 concurrent clients (local Docker): **14,109 requests, 0% HTTP failures, ∼500 req/s steady-state, 96 ms p95 latency**; 11 Pytest integration tests in GitHub Actions CI.

**Recall – Full-Stack AI Research Workspace** | *GitHub* | *Live Demo*
*Python, FastAPI, PostgreSQL, pgvector, SQLAlchemy, Next.js, React, TypeScript, Docker, Ollama*
- Built a FastAPI service that chunks academic PDFs with PyMuPDF and indexes 384-dim embeddings in pgvector.
- Implemented hybrid RAG retrieval combining vector search and BM25 with query expansion, deduplication, and cross-encoder reranking; streamed citation-grounded answers from a local Ollama LLM (gemma3:4b).
- Built a Next.js/TypeScript workspace with PDF evidence viewing, collections, bookmarks, notes, and chat history; deployed a Vercel demo whose pipeline explorer replays recorded retrieval and generation events.

**SpotOn – AI-Powered Fitness Form Analyzer** | HackRU Fall 2026 | *GitHub* | *Live Demo*
*TypeScript, Next.js, React, MediaPipe, Gemini API, ElevenLabs, TimescaleDB, IndexedDB, Vitest, Playwright*
- Co-developed a browser-based squat analyzer on a 3-person team, using MediaPipe pose estimation to track body landmarks for rep counting, joint-angle visualization, skeleton replay, and phase-aligned attempt comparison.
- Built local-first IndexedDB storage with optional TimescaleDB sync; added Gemini feedback and ElevenLabs voice.

## Certifications & Awards

**Certifications**: Google Cybersecurity Certificate; Indian Sign Language Training
**Awards**: Duke of Edinburgh's International Award; Avanti Women's Leadership Program; Model UN Commendation